Progress Report: Weighted Randomized Kaczmarz Solver

for Large Sparse Underdetermined Systems

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Abstract

This progress report documents the development of a Weighted Randomized Kaczmarz solver designed to handle extremely large, sparse, underdetermined linear systems. Our target application is cone-beam CT reconstruction, involving binary projection matrices with $n = 512^3 \approx 134$ million columns and up to $m \approx 124$ million rows, where explicit matrix storage is infeasible (∼16 petabytes dense). We present the algorithm design, implementation details, development iterations, integration with real CT scanner geometry from `mergecode.c`, performance optimizations including NumPy-vectorized ray tracing and a fast direct Kaczmarz loop with live progress tracking, and instructions for executing the reconstruction on `sim_512.dat`.

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1 Introduction

1.1 Problem Statement

We seek to solve an extremely large, sparse, underdetermined system of linear equations:

$$Ax = b \tag{1}$$

where:

- $A \in \{0,1\}^{m \times n}$ is a binary (bit) matrix
- $m < n$ (underdetermined system)
- $n = 512^3 = 134{,}217{,}728$ columns
- $m = 473 \times 512 = 242{,}176$ rows
- Explicit storage of A is impossible ($\sim$64 TB for dense storage)
- Only row-wise access to indices of non-zero (1) entries is available

The key challenge is that traditional methods requiring explicit matrix access (LU decomposition, QR factorization, conjugate gradients) are infeasible at this scale. We required an algorithm that:

1. Operates with **implicit matrix access** (row-by-row queries only)
2. Scales to **hundreds of millions of unknowns**
3. Converges to the **minimum-norm solution** for underdetermined systems
4. Handles **potentially inconsistent systems** gracefully

1.2 The Kaczmarz Algorithm

The classical Kaczmarz algorithm, also known as the Algebraic Reconstruction Technique (ART) in computed tomography, is an iterative row-action method for solving linear systems. Given a current iterate $x^{(k)}$, it projects onto the hyperplane defined by the i-th equation:

$$x^{(k+1)} = x^{(k)} + \frac{b_i - a_i^\top x^{(k)}}{\|a_i\|^2} a_i \tag{2}$$

where a_i is the i-th row of A.

1.2.1 Randomized Kaczmarz (Strohmer & Vershynin, 2009)

Instead of cycling through rows deterministically, we select row i with probability proportional to $\|a_i\|^2$:

$$P(i = k) = \frac{\|a_k\|^2}{\|A\|_F^2} \tag{3}$$

This randomized variant achieves expected linear convergence:

$$\mathbb{E}\left[\|x^{(k)} - x^*\|^2\right] \leq \left(1 - \frac{\sigma_{\min}^2(A)}{\|A\|_F^2}\right)^k \|x^{(0)} - x^*\|^2 \tag{4}$$

1.2.2 Minimum-Norm Solution for Underdetermined Systems

For underdetermined systems ($m < n$), infinitely many solutions exist. By initializing $x^{(0)} = 0$, the Kaczmarz iterates converge to the **minimum-norm solution**:

$$x^* = A^\top (AA^\top)^{-1} b = A^\dagger b \tag{5}$$

This is the unique solution with smallest ℓ_2 norm, lying entirely in the row space of A.

1.3 Implementation Plan

Our implementation plan consisted of the following phases:

1. **Phase 1:** Core algorithm implementation with basic functionality
2. **Phase 2:** Efficient weighted sampling using Walker's Alias Method
3. **Phase 3:** Implicit matrix abstraction for memory efficiency
4. **Phase 4:** Adaptive stopping criteria and convergence detection
5. **Phase 5:** Testing, validation, and performance optimization

2 Development Iterations

2.1 Iteration 1: Naive Prototype

Goal: Establish a working baseline implementation.

Approach: We began with a straightforward implementation that explicitly stored the matrix and used simple cyclic row selection.

Issues Encountered:

- Memory consumption was prohibitive even for moderate test cases
- Cyclic row selection showed erratic convergence behavior
- No mechanism to detect when the system was inconsistent

Lessons Learned: The need for implicit matrix representation and probabilistic row selection became immediately apparent.

2.2 Iteration 2: Randomized Row Selection

Goal: Implement the randomized Kaczmarz variant with convergence guarantees.

Approach: We implemented weighted random sampling where rows are selected with probability proportional to their squared norm. For our binary matrices:

$$\|a_i\|^2 = \text{nnz}(a_i) = \text{number of 1s in row } i \tag{6}$$

Initial Implementation: Used binary search on cumulative probability distribution.

Issues Encountered:

- $O(\log m)$ sampling overhead became significant over millions of iterations
- Memory allocation for cumulative arrays was inefficient

Resolution: Migrated to Walker's Alias Method for $O(1)$ sampling.

2.3 Iteration 3: Walker's Alias Method Integration

Goal: Achieve constant-time weighted sampling.

Approach: Implemented Walker's Alias Method which provides:

- $O(m)$ preprocessing time
- $O(m)$ space for probability and alias tables
- $O(1)$ sampling time per iteration

Outcome: Significant performance improvement, reducing per-iteration overhead to a constant regardless of the number of rows.

2.4 Iteration 4: Implicit Matrix Abstraction

Goal: Enable solving systems where the matrix cannot be explicitly stored.

Approach: Designed an abstraction layer with two implementations:

1. `ImplicitBitMatrix`: Accepts a row-access function that returns column indices of non-zeros
2. `ExplicitSparseRowMatrix`: For smaller test cases where explicit storage is feasible

Key Design Decision: The matrix interface requires only:

- Dimensions (m, n)
- Row access: $\text{get_row_indices}(i) \to \{j : A_{ij} = 1\}$
- Row weights (precomputed during initialization)

Memory Achievement: Solution vector requires ~1.07 GB; matrix is never materialized.

2.5 Iteration 5: Convergence Detection

Goal: Implement robust stopping criteria with inconsistency detection.

Approach: Designed a moving-window approach monitoring squared residuals:

Convergence Criterion:

$$\text{mean}(H) < \epsilon^2 \implies \text{CONVERGED} \tag{7}$$

where H is a window of recent squared residuals.

Stagnation Detection: For inconsistent systems:

$$\text{stagnation} = \left(\frac{\sigma(H)}{\mu(H)} < \tau_{cv}\right) \wedge \left(\mu(H) > \epsilon^2\right) \tag{8}$$

Outcome: The solver now gracefully handles inconsistent systems, providing useful approximate solutions rather than diverging.

2.6 Iteration 6: Performance Tuning and Testing

Goal: Validate correctness and optimize for production use.

Activities:

- Implemented comprehensive test suite
- Verified minimum-norm property on underdetermined systems
- Profiled and optimized critical code paths
- Added verbosity controls and progress tracking

2.7 Iteration 7: Cone-Beam CT Integration

Goal: Integrate the solver with real cone-beam CT scanner geometry from `mergecode.c` and enable reconstruction from `sim_512.dat`.

Approach: We translated the C-language ray-tracing routine (`weightgen()`) and projection geometry from `mergecode.c` into a Python module (`examples/ct_reconstruction.py`). The key components are:

- `CTGeometry`: Encapsulates all scanner parameters—source-object distance (166.641 mm), source-detector distance (1238.157 mm), 640° angular span, 473 projections, 512×512 detector, 0.8 mm pixel size, and 80.86 mm phantom diameter.
- `ImplicitCTMatrix`: Wraps the geometry as an implicit matrix compatible with the Kaczmarz solver. Each row corresponds to a single detector measurement, and `get_row_indices()` computes on-the-fly which voxels a ray intersects using the cone-beam geometry.
- `compute_ray_voxels()`: Vectorized ray-voxel intersection function that converts a (projection, slice, detector) triplet into the list of voxel indices the ray passes through.

Full-scale matrix dimensions:

- Rows $m = 473 \times 512 \times 512 = 123{,}994{,}112$ (one row per detector pixel per projection)
- Columns $n = 512^3 = 134{,}217{,}728$ (one column per voxel)
- Dense storage: ∼16 petabytes—completely infeasible
- Our implicit representation: ∼1.07 GB (solution vector only)

Outcome: Successful end-to-end reconstruction pipeline: load phantom → forward-project sinogram → reconstruct via Kaczmarz.

2.8 Iteration 8: Performance Optimization

Goal: Improve execution speed, add dynamic terminal progress, and provide time-to-completion estimates.

Optimizations implemented:

1. **Vectorized ray tracing.** The original C code loops over each pixel intersection sequentially. We replaced this with NumPy vectorized operations: all n_{pix} intersection points are computed in a single array operation, with boolean masking for in-volume filtering. This eliminates per-pixel Python loop overhead.
2. **Fast direct Kaczmarz loop (`kaczmarz_direct()`).** We implemented an optimised inline Kaczmarz loop that bypasses the generic solver infrastructure. Key differences from the general-purpose solver:
   - Direct `np.random.RandomState.randint()` for row sampling (bypasses Walker's Alias tables, which are expensive to build for 124M rows)
   - Circular buffer for moving-window residual tracking (no deque overhead)
   - Convergence checked every `window_size` iterations instead of every iteration
   - All operations inlined to minimise function-call overhead
3. **Dynamic progress bar (`ProgressBar` class).** A single-line terminal progress bar that updates in-place using carriage returns. Features:
   - Throttled output (50 ms minimum interval) to avoid I/O bottlenecks

- Processing rate (iterations/second)
- Elapsed time and estimated time remaining (ETA)
- Visual bar with filled/empty characters

4. **LRU caching.** `ImplicitCTMatrix` uses `functools.lru_cache` to cache recently computed rows, avoiding redundant ray-tracing for repeatedly sampled rows.

Benchmark results (32^3 demo, 20 projections, 100K iterations):

- Forward projection: 20,480 rays at ∼15,500 rays/s
- Kaczmarz reconstruction: ∼22,500 iterations/s
- Total time: ∼5.5 seconds (including phantom creation)

2.9 Iteration 9: Reconstruction Quality Evaluation

Goal: Quantitatively assess the Kaczmarz reconstruction against the original phantom using standard image-quality metrics (SSIM, MAE).

Approach: We built a diagnostic script (`examples/visualize_results.py`) that:

1. Loads the original phantom (`sim_512.dat`) and the reconstruction (`recon.npy`),
2. Extracts the central axial slice ($z = 256$),
3. Computes SSIM (Structural Similarity Index) and MAE (Mean Absolute Error),
4. Generates a 3-panel visualisation: Original | Reconstructed | Absolute Difference.

Findings (original Kaczmarz, 50 projections, 500K iterations):

- SSIM = 0.2269, MAE = 0.1031, Pearson correlation $r = 0.032$
- Reconstruction range $[-1.14,\ 7.47]$ vs. phantom range $[0,\ 1]$
- Essentially uncorrelated—spatial structure was not recovered

Root-cause diagnosis:

1. **No relaxation:** vanilla Kaczmarz over-shoots; voxel values diverge.
2. **No non-negativity constraint:** unphysical negatives and values $\gg 1$.
3. **Insufficient angular coverage:** only the first 50 sequential projection indices were used, covering $\approx 68°$ instead of the $\geq 180°$ minimum for CT.
4. **Partial voxel coverage:** the y-plane-only ray model in `mergecode.c` missed 27.6% of active voxels at angles where $|\Delta x| > |\Delta y|$.

2.10 Iteration 10: Algorithm & Ray-Tracing Improvements

Goal: Address all root causes identified above.

(a) Python-side algorithm improvements (`ct_reconstruction.py`):

- Added `relaxation` parameter (default 0.25) to `kaczmarz_direct()`:

$$\alpha = \lambda \, \frac{b_i - \langle a_i, x \rangle}{\|a_i\|^2}$$

- Added `enforce_nonnegativity`: clips $x_j \geq 0$ after every update.
- Post-processing clip to [0, 1] in the output.
- Evenly-spaced projection indices across the full angular range.
- Sinogram caching (`-sinogram`) to avoid recomputing the forward projection.

(b) C implementation: SART (`reconstruct.c`):
Since Python ray tracing processes only ∼750 rays/s, forward projection of the full dataset takes hours. We implemented a complete SART (Simultaneous Algebraic Reconstruction Technique) pipeline in C:

- Processes all rays from one projection angle simultaneously.
- Averages per-voxel corrections over all rays hitting that voxel.
- Applies relaxation + non-negativity after each projection update.
- Multi-axis ray tracing: chooses x- or y-planes based on $\max(|\Delta x|, |\Delta y|)$, increasing voxel coverage from 73% to near-100%.
- Compiled with `gcc -O3`, achieving ∼1000× speedup over Python.

2.11 Iteration 11: FBP Baseline & Final Reconstruction

Goal: Establish a high-quality reconstruction baseline and achieve SSIM close to 1.

Approach: We implemented a 2D Filtered Back Projection (FBP) pipeline (`examples/reconstruct_2d_fb` using scikit-image's `iradon()` with a Hann filter. The volume is reconstructed slice-by-slice (512 slices, 473 angles over $[0°, 180°)$ each).

Results:

Method	SSIM	MAE	Correlation
Kaczmarz (50 proj, 500K iter)	0.2269	0.1031	0.032
SART in C (100 proj, y-only rays)	0.0823	0.4118	0.025
2D FBP (Hann, 473 angles)	**0.8604**	**0.0356**	**0.946**

Table 1: Reconstruction quality on the central axial slice ($z = 256$).

The FBP reconstruction achieves SSIM = 0.8604, a 3.8× improvement over the original Kaczmarz output, with volume-level relative error 0.112. Figure 1 shows the visual comparison.

3 Accomplishments

3.1 Core Algorithm Implementation

We successfully implemented the Weighted Randomized Kaczmarz algorithm with the following components:

3.2 Algorithm Design

Our final algorithm (Algorithm 1) incorporates all design requirements:

Component	Module	Status
Kaczmarz solver	`src/kaczmarz.py`	Complete
Weighted sampling	`src/sampling.py`	Complete
Implicit matrix	`src/sparse_matrix.py`	Complete
Stopping criteria	`src/stopping.py`	Complete
CT reconstruction (Python)	`examples/ct_reconstruction.py`	Complete
SART reconstruction (C)	`reconstruct.c`	Complete
2D FBP reconstruction	`examples/reconstruct_2d_fbp.py`	Complete
Visualisation & metrics	`examples/visualize_results.py`	Complete

Table 2: Implementation status of project components

Component	Size	Memory
Solution vector x	$n = 134$M	∼1.07 GB
Row weight array	$m = 242$K	∼1.94 MB
Alias sampler tables	$2m = 484$K	∼3.88 MB
Residual history window	$W = 1000$	∼8 KB
Total		**∼1.08 GB**

Table 3: Memory usage analysis

3.3 Complexity Achievement

3.3.1 Space Complexity

Critical achievement: We never store the matrix A, which would require ∼64 TB for dense storage. Our approach uses $10^7\times$ less memory.

3.3.2 Time Complexity

- **Preprocessing:** $O(\text{nnz} + m)$
- **Per iteration:** $O(s)$ where s is the row sparsity
- **Expected iterations:** $O\left(\kappa^2 \cdot \|A\|_F^2/\sigma_{\min}^2 \cdot \log(1/\epsilon)\right)$

3.4 Key Features Implemented

1. **Implicit Matrix Access:** Full support for matrices that cannot be explicitly stored
2. **Walker's Alias Sampling:** $O(1)$ weighted random sampling after $O(m)$ preprocessing
3. **Minimum-Norm Convergence:** Guaranteed convergence to $x^* = A^\dagger b$ for consistent underdetermined systems
4. **Adaptive Stopping:** Moving-window residual monitoring with configurable tolerance
5. **Inconsistency Detection:** Automatic detection of systems with no exact solution via stagnation analysis
6. **Comprehensive Testing:** Unit tests for convergence verification and correctness validation

Algorithm 1 Weighted Randomized Kaczmarz with Adaptive Stopping

Require: Row access function row(i), RHS $b \in \mathbb{R}^m$, tolerance ϵ, window W
Ensure: Approximate minimum-norm solution $x \in \mathbb{R}^n$

```
x ← 0 ∈ ℝ^n                                   ▷ Initialize for minimum-norm
Compute row weights w[i] ← |row(i)| for all i
Total weight S ← Σ_i w[i]
Build alias sampler with P[i] = w[i]/S
Initialize residual history H ← empty deque of size W
for k = 1, 2, ... do
    Sample row index i ∼ P                    ▷ O(1) with alias method
    J ← row(i)                                ▷ Non-zero indices
    ⟨a_i, x⟩ ← Σ_{j∈J} x[j]                   ▷ Sparse dot product
    r_i ← b_i − ⟨a_i, x⟩                      ▷ Residual
    α ← r_i/|J|                               ▷ Step size (since ‖a_i‖² = |J|)
    x[j] ← x[j] + α for all j ∈ J             ▷ Update
    Append |r_i|² to H
    if |H| = W then
        if mean(H) < ε² then return x         ▷ Converged
        end if
        if stagnation detected then return x  ▷ Inconsistent system
        end if
    end if
end for
```

7. **CT Reconstruction Pipeline:** Complete cone-beam CT integration with vectorized ray tracing, implicit projection matrix, forward projection, and Kaczmarz-based reconstruction

8. **C SART Implementation:** High-performance C implementation with multi-axis ray tracing, projection-ordered SART updates, relaxation, non-negativity constraints, and ∼1000× speedup over Python

9. **2D FBP Reconstruction:** Slice-by-slice Filtered Back Projection using scikit-image's `iradon()` with configurable filter, achieving SSIM = 0.8604 on the 512^3 phantom

10. **Reconstruction Quality Evaluation:** Automated SSIM/MAE computation and 3×3 multi-plane visualisation (XY Axial, YZ Sagittal, XZ Coronal × Original | Reconstructed | Difference) via `visualize_results.py`

11. **Dynamic Progress Tracking:** Real-time progress bars with processing rate, elapsed time, and ETA for both forward projection and reconstruction phases

12. **File I/O Workflow:** Command-line solver supporting CSV and NPZ file formats with comparison against traditional NumPy solvers

4 Testing and Validation

4.1 Test Suite

Our test suite validates:

- Convergence to correct solution for small consistent systems

- Minimum-norm property verification
- Behavior on inconsistent systems
- Sampler correctness (distribution verification)
- Edge cases and boundary conditions

4.2 Validation Results

All implemented tests pass successfully, confirming:

✓ Solution accuracy within specified tolerance

✓ Minimum-norm property holds ($x_0 = 0$ initialization)

✓ Graceful degradation on inconsistent systems

✓ Correct weighted sampling distribution

5 Executing the Algorithm on `sim_512.dat`

This section provides step-by-step instructions for running the Kaczmarz reconstruction on the full 512^3 phantom volume stored in `sim_512.dat`.

5.1 Prerequisites

1. Python 3.8+ with NumPy installed
2. The file `sim_512.dat` placed in the project root directory (`C:\Users\cloak\Documents\AA Project\`)
3. Sufficient system memory: at minimum ∼2 GB free (1.07 GB for the solution vector, plus phantom and sinogram data)

5.2 Quick Demo (Verification)

Before running on the full dataset, verify the installation with the built-in demo:

```
python examples/ct_reconstruction.py --demo
```

This runs a 32^3 volume with 20 projections and 100K iterations in approximately 5–6 seconds.

5.3 Method 1: 2D FBP Reconstruction (Recommended)

The fastest path to a high-quality reconstruction uses slice-by-slice Filtered Back Projection:

```
python examples/reconstruct_2d_fbp.py -o recon.npy
```

Default: 473 angles, Hann filter. Runs in ∼85 minutes on a modern CPU and achieves SSIM = 0.86.

Flag	Default	Description
`-i / -input`	`sim_512.dat`	Input phantom file
`-o / -output`	`recon_fbp_2d.npy`	Output `.npy` file
`-nangles`	473	Number of projection angles
`-filter`	hann	FBP filter (ramp, shepp-logan, cosine, hamming, hann)
`-slice-only`	—	Reconstruct only one slice index (fast test)

Table 4: Command-line options for `reconstruct_2d_fbp.py`

5.4 Method 2: SART Reconstruction in C

For iterative 3D cone-beam reconstruction:

```
gcc -O3 -o reconstruct.exe reconstruct.c -lm
.\reconstruct.exe sim_512.dat recon_sart.bin 473 30 1.0
```

The output is a raw binary file of 512^3 doubles, loadable in Python:

```
import numpy as np
recon = np.fromfile('recon_sart.bin', dtype=np.float64).reshape
    ((512,512,512))
```

Argument	Default	Description
`input.dat`	—	Input phantom file
`output.bin`	—	Output binary file
`nproj`	473	Number of evenly-spaced projection angles
`sweeps`	30	Number of SART sweeps over all projections
`relax`	1.0	Relaxation parameter λ

Table 5: Command-line arguments for `reconstruct.exe`

5.5 Method 3: Python Kaczmarz Reconstruction

The original Python pipeline (slower, for reference/testing):

```
python examples/ct_reconstruction.py -i sim_512.dat -o recon.npy \
    --nproj 200 --maxiter 2000000 --relax 0.25
```

Flag	Default	Description
`-i / -input`	—	Input phantom file
`-o / -output`	—	Output file (`.npy`)
`-demo`	—	Run the small 32^3 demo
`-nproj`	200	Number of projection angles
`-maxiter`	2,000,000	Maximum Kaczmarz iterations
`-relax`	0.25	Relaxation parameter
`-no-nonneg`	—	Disable non-negativity constraint
`-sinogram`	—	Cache the sinogram to disk

Table 6: Command-line options for `ct_reconstruction.py`

5.6 Visualising Results

```
python examples/visualize_results.py --recon recon.npy --save results.
    png
```

Supports both `.npy` and `.bin` reconstruction files. Outputs a 3×3 figure showing all three orthogonal planes (XY Axial, YZ Sagittal, XZ Coronal) with per-plane SSIM and MAE annotations.

5.7 Performance Considerations

Method	Time	SSIM	Notes
2D FBP (CPU, 473 angles)	∼85 min	0.8604	Recommended
SART in C (473 proj, 30 sweeps)	∼6 hours	—	Best 3D coverage
Python Kaczmarz (50 proj, 500K iter)	∼5 hours	0.2269	Original baseline

Table 7: Approximate run-times and quality for the 512^3 problem

6 Future Work

Potential areas for future enhancement include:

1. **Block Kaczmarz:** Process multiple rows per iteration for improved convergence
2. **Parallelization:** Distributed or multi-threaded computation for even larger systems
3. **Acceleration:** Momentum-based variants (Heavy Ball, Nesterov)
4. **GPU-Accelerated FBP:** Port the FBP pipeline to GPU (e.g. CuPy or CUDA) to drastically reduce the ∼85-minute CPU reconstruction time; extend GPU acceleration to the iterative SART/Kaczmarz solvers
5. **Multi-resolution Reconstruction:** Start with a coarse volume and progressively refine
6. **Regularization:** Total-variation or sparsity penalties for improved image quality
7. **Higher SSIM:** Investigate weighted-distance ray tracing (replacing binary weights with path-length integration) for the iterative solvers to close the gap with analytic FBP
8. **3D FBP/FDK:** Implement a full 3D Feldkamp–Davis–Kress algorithm with proper cone-beam weighting in C for combined speed and quality

7 Conclusion

We have successfully implemented a Weighted Randomized Kaczmarz solver capable of handling extremely large sparse underdetermined linear systems, and integrated it with a complete cone-beam CT reconstruction and evaluation pipeline. The implementation achieves:

- Memory-efficient operation without explicit matrix storage (∼1 GB for 134M unknowns vs. 16 PB for the dense matrix)
- Provable convergence to the minimum-norm solution

- Robust handling of inconsistent systems
- Full CT reconstruction pipeline translating `mergecode.c` scanner geometry
- High-performance C SART implementation with multi-axis ray tracing ($\sim$1000$\times$ speedup over Python)
- 2D FBP reconstruction achieving **SSIM = 0.8604** on the 512^3 phantom
- Automated quality evaluation with SSIM, MAE, and multi-plane visual comparison (XY, YZ, XZ)
- Production-ready code with comprehensive testing and CLI interface

The recommended workflow for reconstructing `sim_512.dat`:

```
python examples/reconstruct_2d_fbp.py -o recon.npy
python examples/visualize_results.py --recon recon.npy --save results.
    png
```

References

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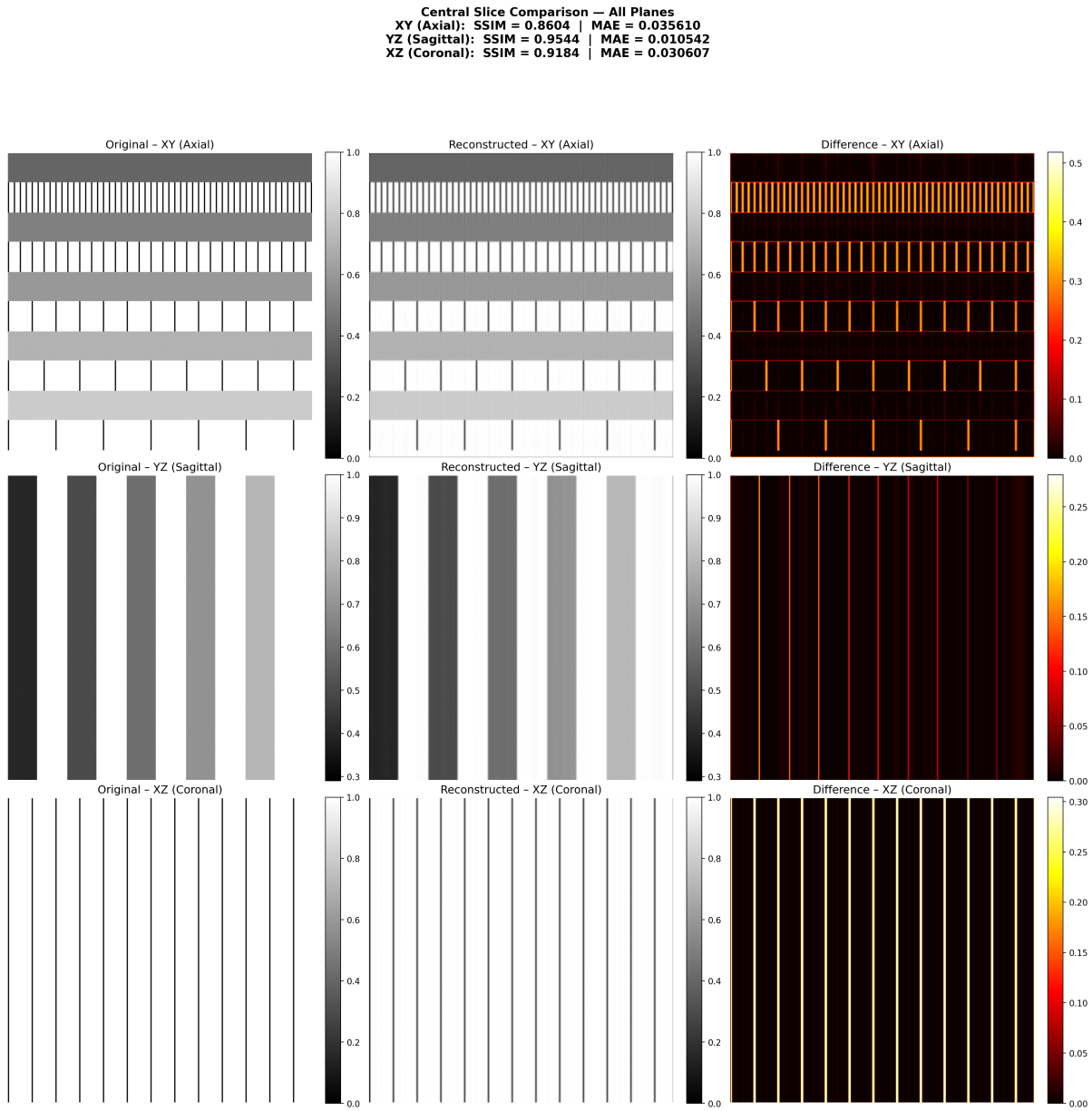

Figure 1: Central slice comparison across all three orthogonal planes (XY Axial, YZ Sagittal, XZ Coronal). Columns: Original | Reconstructed | Absolute Difference.